

But in-house IT systems need periodic maintenance during which it becomes difficult to provide continued service. Cloud computing offers server availability of 99.999%, even during maintenance, to do transactions at any time of the day or night. The hosting of mobile and web apps also ensures better speed for users.

The confidentiality and security of financial information of customers and internal company data are of prime importance. Cloud computing is used to ensure that payments are processed securely without any discrepancies. It is engineered to offer multiple layers of safety and can pinpoint fraudulent transactions in real-time without compromising user experience.

Cloud computing is changing the way consumers interact with banks;

online fund transfer, digital wallets, payment gateways, and secure online payments are among the best examples of cloud computing service. All the basic services such as net banking, mobile banking, KYC, and card management are managed aptly with cloud platforms.

With cloud providers offering pay-as-you-use models that allow their usage and payment only for the capacity being used, adoption of cloud computing can lead to significant savings for banks. Banks can divert those funds to develop other financial products to improve their customers' experience.

Most prominent cloud providers offer sophisticated encryption and end-to-end data protection solutions to help secure their customers' data. Cloud computing architectures offer

speed as well as flexibility, which ultimately make data safer both internally and externally.

When data is secure and can be accessed at any time, anywhere, by both customers and employees without the down-time restrictions imposed by traditional infrastructure, banks can improve operations as well as user experience.

Cloud computing can transform the banking and financial sector, offering support to the industrial revolution 4.0. The technology has been implemented to keep security, regulatory and compliance and business alignment in effect. Experts across the banking sector believe that cloud-based services should be deployed only through trusted cloud service providers.

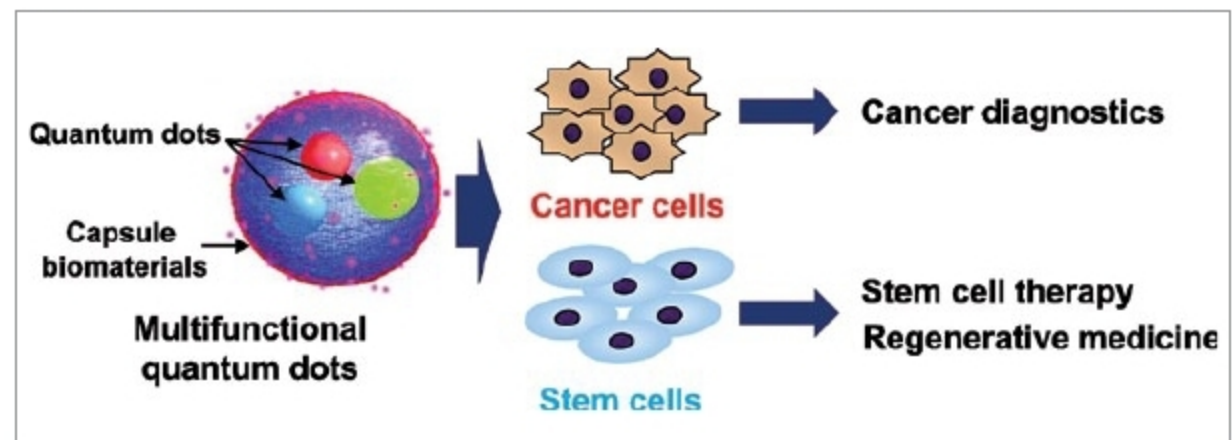
—Deepshikha Shukla

3 QUANTUM DOTS TRANSFORMING THE MEDICAL FIELD

Quantum dots (QDs), popularly used as QLED displays in TVs, are now playing a crucial role in the development and transformation of the medical field. QDs are now being used in advanced and intelligent applications like bio-imaging, drug delivery, sensing, and therapy

Nanotechnology, the branch of science and engineering that deals with the world of nanometer-sized particles, is rapidly growing due to its potential to solve some of the most pressing problems in the world like hunger and serious diseases such as cancer.

An important topic of research in nanotechnology, quantum dots (QDs) that find their popular use as QLED displays to light up TVs have a much wider scope beyond that. QDs are nanoscale crystals made of pure metal or semiconductor alloys with special optic and electronic properties compared to larger particles due to quantum mechanics. Characteristics of these particles include good photostability, high brightness, high quantum yield, narrow and symmetrical emission, broad excitation, and so on. These advantages make them suitable for advanced and intelligent applications like bio-imaging, drug delivery, sensing, and therapy. This is



(Credit: www.sciencedirect.com)

especially beneficial for diseases hard to diagnose or treat.

When ultraviolet rays hit these dots, they can emit light of different colours based on their size. QDs have tunable optical properties, which make them useful as fluorescent dyes for deep-tissue imaging typically used in animal models since most conventional organic dyes are not capable of near-infrared (NIR) emissions. Visible light, with a wavelength range of about 400-700nm, is unsuitable

for transmission through biological tissues but QDs can make use of the NIR optical window. This is done by controlling the size of the QDs.

Development of theranostics platforms for simultaneous sensing, imaging and therapy is also increasing. In theranostics, specific targeted therapy is delivered to a patient based on specific targeted diagnostic images and tests. QDs can behave as nano-carriers for drug delivery or fluorescent labels for a theranostic approach in therapy.